Spin-Orbit Coupled BEC: BdG theory

Eq. (2) in Li2013 states the single-particle hamiltonian is

$$h_0 = \frac{1}{2}(p_x - k_0\sigma_z)^2 + \frac{\Omega}{2}\sigma_x, \tag{1}$$

$$= \frac{1}{2}\begin{bmatrix} p_x^2 - 2k_0p_x + k_0^2 & \Omega \\ \Omega & p_x^2 + 2k_0p_x + k_0^2 \end{bmatrix}. \tag{2}$$

The non-linear interaction term in the GPE is

$$V = \begin{bmatrix} g\left|\psi_\uparrow\right|^2 + g_{\uparrow\downarrow}\left|\psi_\downarrow\right|^2 & 0 \\ 0 & g\left|\psi_\downarrow\right|^2 + g_{\uparrow\downarrow}\left|\psi_\uparrow\right|^2 \end{bmatrix}. \tag{3}$$

From Martone's thesis, the ground state in the stripe phase is given by the spinor state:

$$\Psi_0(x) = \begin{pmatrix} \psi_{0\uparrow}(x) \\ \psi_{0\downarrow}(x) \end{pmatrix} = \sqrt{n}\sum_K \begin{pmatrix} a_{K-k_1} \\ -b_{K-k_1} \end{pmatrix} e^{i(K-k_1)x}, \tag{4}$$

which is a solution to the time-independent GPE, so

$$\mu\Psi_0 = \mathcal{L}\Psi_0, \tag{5}$$

where $\mathcal{L} = h_0 + V$.

The two-component perturbation is

$$\delta(x,t) = \begin{pmatrix} \delta_\uparrow \\ \delta_\downarrow \end{pmatrix} = \begin{pmatrix} u_{q\uparrow}(x) \\ u_{q\downarrow}(x) \end{pmatrix} e^{-i\omega t} - \begin{pmatrix} v^*_{q\uparrow}(x) \\ v^*_{q\downarrow}(x) \end{pmatrix} e^{i\omega t}. \tag{6}$$

Due to the periodic stucture, expand $u_{q\uparrow,\downarrow}(x)$ and $v_{q\uparrow,\downarrow}(x)$ in Bloch form (Eqs. (5.3) and (5.4) in Martone's thesis):

$$u_{q\uparrow,\downarrow}(x) = \sum_K U_{q\uparrow\downarrow K} e^{iqx} e^{i(K-k_1)x}, \tag{7}$$

$$v_{q\uparrow,\downarrow}(x) = \sum_K V_{q\uparrow\downarrow K} e^{iqx} e^{-i(K-k_1)x}. \tag{8}$$

The spinor wavefunction of excitations in the stripe phase is thus

$$\Psi(x,t) = \begin{pmatrix}\psi_\uparrow\\ \psi_\downarrow\end{pmatrix} = e^{-i\mu t}\left[\Psi_0(x) + \delta(x,t)\right] \tag{9}$$

$$= e^{-i\mu t}\begin{bmatrix}\psi_{0\uparrow} + \delta_\uparrow\\ \psi_{0\downarrow} + \delta_\downarrow\end{bmatrix} \tag{10}$$

The one-dimension time-dependent GPE equation is

$$i\frac{\partial \Psi}{\partial t} = \mathcal{L}\Psi. \tag{11}$$

Substituting Eq. (10) into Eq. (11) results in the linearised GPE:

$$i\frac{\partial \delta}{\partial t} = (h_0 - \mu)\delta + \begin{bmatrix}2g\left|\psi_{0\uparrow}\right|^2 + g_{\uparrow\downarrow}\left|\psi_{0\downarrow}\right|^2 & g_{\uparrow\downarrow}\psi_{0\uparrow}\psi_{0\downarrow}^*\\ g_{\uparrow\downarrow}\psi_{0\uparrow}^*\psi_{0\downarrow} & 2g\left|\psi_{0\downarrow}\right|^2 + g_{\uparrow\downarrow}\left|\psi_{0\uparrow}\right|^2\end{bmatrix}\delta + \begin{bmatrix}g\psi_{0\uparrow}^2 & g_{\uparrow\downarrow}\psi_{0\uparrow}\psi_{0\downarrow}\\ g_{\uparrow\downarrow}\psi_{0\uparrow}\psi_{0\downarrow} & g\psi_{0\downarrow}^2\end{bmatrix}\delta^*, \tag{12}$$

$$= (\mathcal{L} - \mu)\delta + \begin{bmatrix}g\left|\psi_{0\uparrow}\right|^2 & g_{\uparrow\downarrow}\psi_{0\uparrow}\psi_{0\downarrow}^*\\ g_{\uparrow\downarrow}\psi_{0\uparrow}^*\psi_{0\downarrow} & g\left|\psi_{0\downarrow}\right|^2\end{bmatrix}\delta + \begin{bmatrix}g\psi_{0\uparrow}^2 & g_{\uparrow\downarrow}\psi_{0\uparrow}\psi_{0\downarrow}\\ g_{\uparrow\downarrow}\psi_{0\uparrow}\psi_{0\downarrow} & g\psi_{0\downarrow}^2\end{bmatrix}\delta^*, \tag{13}$$

$$= (\mathcal{L} - \mu + X)\delta + Y\delta^*. \tag{14}$$

Then substituting Eq. (6) into Eq. (14) gives the eigenvalue equation:

$$\omega_{nq}\vec{w}_{nq} = H_{BdG}^{(q)}\vec{w}_{nq} \tag{15}$$

where

$$\vec{w}_{nq} = \begin{pmatrix}u_{nq\uparrow}\\ u_{nq\downarrow}\\ v_{nq\uparrow}\\ v_{nq\downarrow}\end{pmatrix}, \tag{16}$$

and

$$H_{BdG}^{(q)} = \begin{bmatrix}\mathcal{L}^{(q)} - \mu + X^{(q)} & -Y^{(q)}\\ Y^{(-q)*} & -(\mathcal{L}^{(-q)} - \mu + X^{(-q)})^*\end{bmatrix}. \tag{17}$$